



EXAM PAPERS PRACTICE

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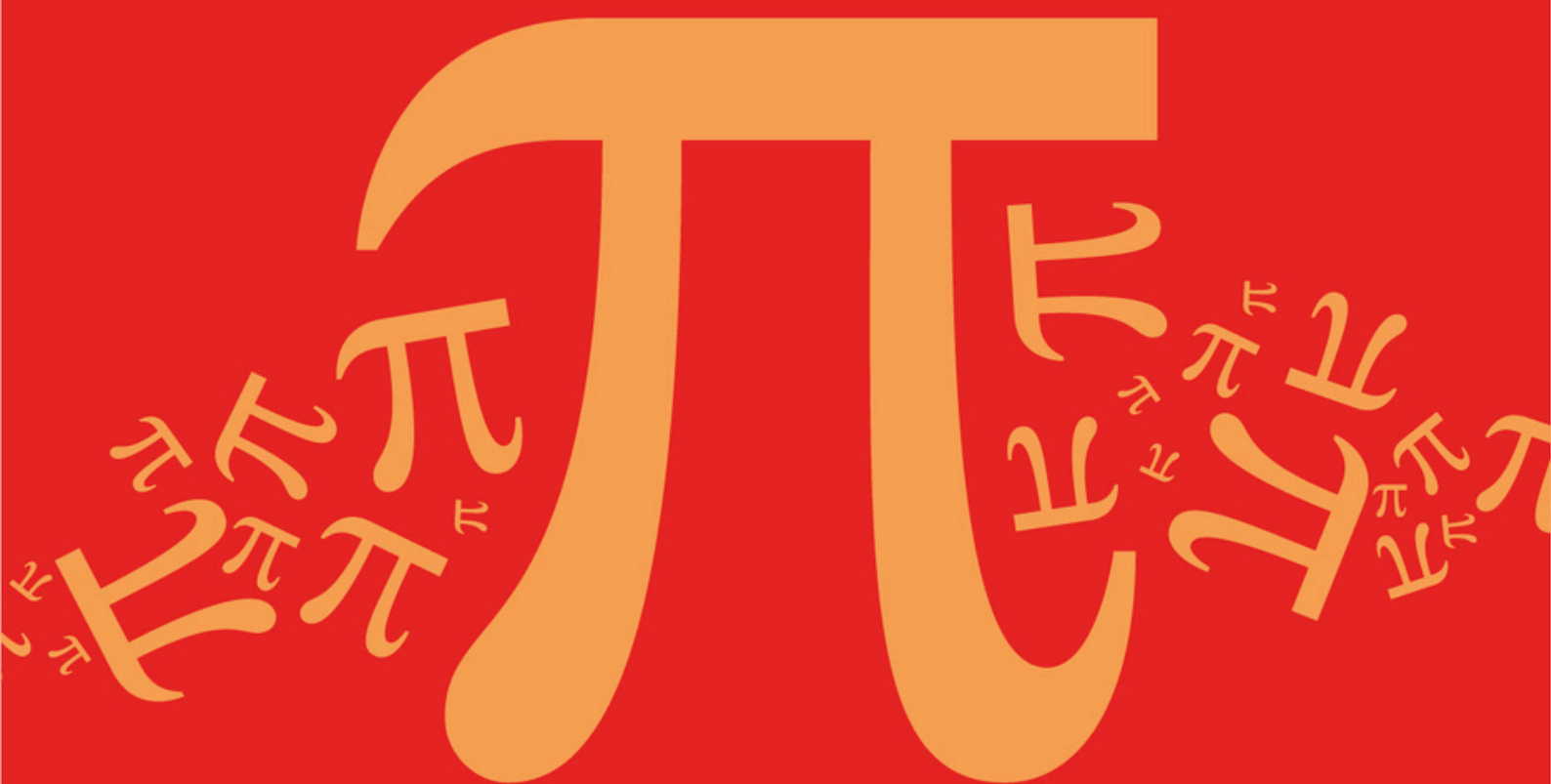
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AQA A Level Further Mathematics (7367)



Topic

D: Further Algebra and Functions

Sub topic

D7: Evaluate Limits Using Maclaurin

Topic Questions

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Question Paper

Topic

D: Further Algebra and Functions

Sub topic

D7: Evaluate Limits Using Maclaurin

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**To be used by all students and teachers for
AQA A Level Further Mathematics (7367)
for exams from 2025 onwards.**

Students and teachers of other boards may also find this useful

Name : _____

Date : _____

Time : _____

Total Marks Available : _____

Total Marks Received : _____

Q1.

- (a) Write down the expansion of $\sin 2x$ in ascending powers of x up to and including the term in x^5 .

(1)

- (b) Show that, for some value of k ,

$$\lim_{x \rightarrow 0} \left[\frac{2x - \sin 2x}{x^2 \ln(1 + kx)} \right] = 16$$

and state this value of k .

(4)

(Total 5 marks)

Q2.

- (a) Write down the expansions in ascending powers of x up to and including the term in x^3 of:

(i) $\cos x + \sin x$;

(1)

(ii) $\ln(1 + 3x)$.

(1)

- (b) It is given that $y = e^{\tan x}$.

(i) Find $\frac{dy}{dx}$ and show that $\frac{d^2 y}{dx^2} = (1 + \tan x)^2 \frac{dy}{dx}$

(5)

(ii) Find the value of $\frac{d^3 y}{dx^3}$ when $x = 0$.

(2)

- (iii) Hence, by using Maclaurin's theorem, show that the first four terms in the expansion, in ascending powers of x , of $e^{\tan x}$ are

$$1 + x + \frac{1}{2}x^2 + \frac{1}{2}x^3$$

(2)

- (c) Find

$$\lim_{x \rightarrow 0} \left[\frac{e^{\tan x} - (\cos x + \sin x)}{x \ln(1 + 3x)} \right]$$

(3)

(Total 14 marks)

Q3.

- (a) Given that $y = \ln(1 + 2 \tan x)$, find $\frac{dy}{dx}$ and $\frac{d^2y}{dx^2}$.

(You may leave your expression for $\frac{d^2y}{dx^2}$ unsimplified.)

(4)

- (b) Hence, using Maclaurin's theorem, find the first two non-zero terms in the expansion, in ascending powers of x , of $\ln(1 + 2 \tan x)$.

(2)

- (c) Find

$$\lim_{x \rightarrow 0} \left[\frac{\ln(1 + 2 \tan x)}{\ln(1 - x)} \right]$$

(4)

(Total 10 marks)

Q4.

- (a) Given that $y = \ln(4 + 3x)$, find $\frac{dy}{dx}$ and $\frac{d^2y}{dx^2}$.

(3)

- (b) Hence, by using Maclaurin's theorem, find the first three terms in the expansion, in ascending powers of x , of $\ln(4 + 3x)$.

(2)

- (c) Write down the first three terms in the expansion, in ascending powers of x , of $\ln(4 - 3x)$.

(1)

- (d) Show that, for small values of x ,

$$\ln \left(\frac{4 + 3x}{4 - 3x} \right) \approx \frac{3}{2}x$$

(2)

(Total 8 marks)

Q5.

- (a) Write down the expansion of $\sin 3x$ in ascending powers of x up to and including the term in x^3 .

(1)

- (b) Find

$$\lim_{x \rightarrow 0} \left[\frac{3x \cos 2x - \sin 3x}{5x^3} \right]$$

(4)

(Total 5 marks)

Q6.

- (a) Write down the expansion of $\cos 4x$ in ascending powers of x up to and including the term in x^4 . Give your answer in its simplest form.

(2)

- (b) (i) Given that $y = \ln(2 - e^x)$, find $\frac{dy}{dx}$, $\frac{d^2y}{dx^2}$ and $\frac{d^3y}{dx^3}$.

(You may leave your expression for $\frac{d^3y}{dx^3}$ unsimplified.)

(6)

- (ii) Hence, by using Maclaurin's theorem, show that the first three non-zero terms in the expansion, in ascending powers of x , of $\ln(2 - e^x)$ are

$$-x - x^2 - x^3$$

(2)

- (c) Find

$$\lim_{x \rightarrow 0} \left[\frac{x \ln(2 - e^x)}{1 - \cos 4x} \right]$$

(3)

(Total 13 marks)

Q7.

The function f is defined by $f(x) = e^{2x}(1 + 3x)^{-\frac{2}{3}}$.

- (a) (i) Use the series expansion for e^x to write down the first four terms in the series expansion of e^{2x} .

(2)

- (ii) Use the binomial series expansion of $(1 + 3x)^{-\frac{2}{3}}$ and your answer to part (a)(i) to show that the first three non-zero terms in the series expansion of $f(x)$ are $1 + 3x^2 - 6x^3$.

(5)

- (b) (i) Given that $y = \ln(1 + 2 \sin x)$, find $\frac{d^2y}{dx^2}$.

(4)

- (ii) By using Maclaurin's theorem, show that, for small values of x ,

$$\ln(1 + 2 \sin x) \approx 2x - 2x^2$$

(2)

- (c) Find

$$\lim_{x \rightarrow 0} \frac{1 - f(x)}{x \ln(1 + 2 \sin x)}$$

(3)

(Total 16 marks)

Q8.

The function f is defined by

$$f(x) = (9 + \tan x)^{\frac{1}{2}}$$

- (a) (i) Find $f''(x)$.

(4)

- (ii) By using Maclaurin's theorem, show that, for small values of x ,

$$(9 + \tan x)^{\frac{1}{2}} \approx 3 + \frac{x}{6} - \frac{x^2}{216}$$

(3)

- (b) Find

$$\lim_{x \rightarrow 0} \left[\frac{f(x) - 3}{\sin 3x} \right]$$

(3)

(Total 10 marks)

Q9.

- (a) (i) Write down the expansion of $\ln(1 + 2x)$ in ascending powers of x up to and including the term in x^3 .

(2)

- (ii) State the range of values of x for which this expansion is valid.

(1)

- (b) (i) Given that $y = \ln \cos x$, find $\frac{dy}{dx}$, $\frac{d^2y}{dx^2}$ and $\frac{d^3y}{dx^3}$.

(4)

- (ii) Find the value of $\frac{d^4y}{dx^4}$ when $x = 0$.

(3)

- (iii) Hence, by using Maclaurin's theorem, show that the first two non-zero terms in the expansion, in ascending powers of x , of $\ln \cos x$ are

$$-\frac{x^2}{2} - \frac{x^4}{12}$$

(2)

- (c) Find

$$\lim_{x \rightarrow 0} \left[\frac{x \ln(1+2x)}{x^2 - \ln \cos x} \right]$$

(3)

(Total 15 marks)

Q10.

- (a) Write down the expansion of $\sin 2x$ in ascending powers of x up to and including the term in x^3 .

(1)

- (b) (i) Given that $y = \sqrt{3 + e^x}$, find the values of $\frac{dy}{dx}$ and $\frac{d^2y}{dx^2}$ when $x = 0$.

(5)

- (ii) Using Maclaurin's theorem, show that, for small values of x ,

$$\sqrt{3 + e^x} \approx 2 + \frac{1}{4}x + \frac{7}{64}x^2$$

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(2)

- (c) Find

$$\lim_{x \rightarrow 0} \left[\frac{\sqrt{3 + e^x} - 2}{\sin 2x} \right]$$

(3)

(Total 11 marks)

Q11.

The function f is defined by $f(x) = (1 + 2x)^{\frac{1}{2}}$.

- (a) (i) Find $f'''(x)$.

(4)

- (ii) Using Maclaurin's theorem, show that, for small values of x ,

$$f(x) \approx 1 + x - \frac{1}{2}x^2 + \frac{1}{2}x^3$$

(4)

- (b) Use the expansion of e^x together with the result in part (a)(ii) to show that, for small values of x ,

$$e^x (1+2x)^{\frac{1}{2}} \approx 1 + 2x + x^2 + kx^3$$

where k is a rational number to be found.

(3)

- (c) Write down the first four terms in the expansion, in ascending powers of x , of e^{2x} .

(1)

- (d) Find

$$\lim_{x \rightarrow 0} \frac{e^x (1+2x)^{\frac{1}{2}} - e^{2x}}{1 - \cos x}$$

(4)

(Total 16 marks)

Q12.

- (a) The function f is defined by

$$f(x) = \ln(1 + e^x)$$

Use Maclaurin's theorem to show that when $f(x)$ is expanded in ascending powers of x :

- (i) the first three terms are

$$\ln 2 + \frac{1}{2}x + \frac{1}{8}x^2$$

(6)

- (ii) the coefficient of x^3 is zero.

(3)

- (b) Hence write down the first two non-zero terms in the expansion, in ascending

powers of x , of $\ln\left(\frac{1+e^x}{2}\right)$.

(1)

- (c) Use the series expansion

$$\ln(1+x) = x - \frac{1}{2}x^2 + \frac{1}{3}x^3 - \dots$$

to write down the first three terms in the expansion, in ascending powers of x , of

$$\ln\left(1 - \frac{x}{2}\right)$$

(1)

- (d) Use your answers to parts (b) and (c) to find

$$\lim_{x \rightarrow 0} \left[\frac{\ln \left(\frac{1+e^x}{2} \right) + \ln \left(1 - \frac{x}{2} \right)}{x - \sin x} \right]$$

(4)

(Total 15 marks)

Q13.

- (a) Use the series expansion

$$\ln(1+x) = x - \frac{1}{2}x^2 + \frac{1}{3}x^3 - \frac{1}{4}x^4 + \dots$$

to write down the first four terms in the expansion, in ascending powers of x , of $\ln(1-x)$.

(1)

- (b) The function f is defined by

$$f(x) = e^{\sin x}$$

Use Maclaurin's theorem to show that when $f(x)$ is expanded in ascending powers of x :

- (i) the first three terms are

$$1 + x + \frac{1}{2}x^2$$

(6)

- (ii) the coefficient of x^3 is zero.

(3)

- (c) Find

$$\lim_{x \rightarrow 0} \frac{e^{\sin x} - 1 + \ln(1-x)}{x^2 \sin x}$$

(4)

(Total 14 marks)

Q14.

- (a) (i) Write down the first three terms of the binomial expansion of $(1+y)^{-1}$, in ascending powers of y .

(1)

- (ii) By using the expansion

$$\cos x = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \dots$$

and your answer to part (a)(i), or otherwise, show that the first three non-zero terms in the expansion, in ascending powers of x , of $\sec x$ are

$$1 + \frac{x^2}{2} + \frac{5x^4}{24}$$

(5)

- (b) By using Maclaurin's theorem, or otherwise, show that the first two non-zero terms in the expansion, in ascending powers of x , of $\tan x$ are

$$x + \frac{x^3}{3}$$

(3)

- (c) Hence find $\lim_{x \rightarrow 0} \left(\frac{x \tan 2x}{\sec x - 1} \right)$.

(4)

(Total 13 marks)